

Kanad Motiwale

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Education

Northeastern University

Master of Science in Computer Science (GPA: 3.92/4.0)

Sep 2025 – Dec 2027

Boston, MA

Ramrao Adik Institute of Technology, University of Mumbai

Bachelor of Engineering in Electronics & Telecommunication (CGPI: 8.72/10)

Aug 2019 – May 2023

Mumbai, India

Technical Skills

Languages: Python, JavaScript (ES6+), TypeScript, Java, C++, SQL, R, HTML5, CSS3, Bash

Frameworks & Libraries: React, Next.js, Node.js, Express.js, FastAPI, pandas, NumPy, Drizzle ORM, Selenium, OpenCV

AI / ML: LLMs, RAG, Agentic AI, OpenAI API, FAISS, Pydantic AI, NLP, Embeddings, Vector Databases, Prompt Engineering

Data Engineering: ETL/ELT, Dimensional Modeling, Data Warehousing, PostgreSQL, MySQL, SQLite, Query Optimization

Cloud & DevOps: AWS (Lambda, EC2, S3), Docker, GitHub Actions, Azure DevOps, CI/CD, Git, Linux

Practices: REST APIs, Microservices, Distributed Systems, MVC, TDD, Unit/Integration Testing, Agile/Scrum, System Design

Experience

LTM

Dec 2023 – Aug 2025

Software Engineer

Mumbai, India

- Architected 6+ fault-tolerant distributed backend services and REST APIs in Python and Java on AWS (Lambda, EC2) for Marsh McLennan's Policy & Placement platform, reducing API latency by 22% and database load by 35% for 10,000+ enterprise users.
- Engineered and maintained Python ETL/ELT data pipelines with anomaly detection and statistical monitoring; resolved 20+ high-severity production incidents via root cause analysis and performance profiling, cutting incident resolution time by 45%.
- Designed and executed 100+ unit and integration tests using JUnit and TDD within Agile/Scrum workflows; performed systematic debugging and SQL optimization, reducing production defects by 25% and deployment time by 35% across the full SDLC.
- Developed reusable React/TypeScript UI components with clean MVC separation; drove cross-functional code reviews and data-model alignment, accelerating feature delivery and improving overall code quality.

FastFindFirm

Jun 2022 – Sep 2023

Software Engineer Intern

Mumbai, India

- Built scalable backend services and REST APIs in Python and SQL serving 5,000+ users; designed and maintained ETL data pipelines supporting real-time content ingestion, transformation, and processing.
- Implemented ML-based ranking and filtering algorithms using scikit-learn, reducing search latency by 30% and infrastructure costs by 25% through query optimization, caching, and efficient data modeling.
- Owned full-stack SDLC from system design to production; containerized microservices with Docker, automated CI/CD pipelines cutting manual effort by 60%, and drove 30% monthly active user growth through iterative, data-driven feature delivery.

LTM

Feb 2023 – Apr 2023

Software Engineer Intern

Mumbai, India

- Developed and debugged backend modules and REST APIs in Python and Java applying OOP and clean-code principles across a production-scale distributed system environment.
- Authored 40+ unit and integration tests via TDD, reducing API error rates by 18%; performed systematic debugging and root cause analysis to resolve issues across the full SDLC.
- Collaborated with senior engineers in Agile/Scrum ceremonies on code reviews, data-model discussions, and feature delivery; received full-time return offer upon internship completion.

Projects

Lexorion | Python, FastAPI, FAISS, Ollama, Pydantic AI, React, SQLite | [GitHub](#)

Apr 2026

- Built a fully local RAG pipeline using FAISS, Ollama (llama3.2), and nomic-embed-text embeddings with zero cloud costs.
- Designed an ETL pipeline in FastAPI to parse PDFs, chunk text with tiktoken, and index embeddings into FAISS with SQLite.
- Implemented a Pydantic AI multi-agent re-ranking layer; built a React frontend with chat, upload, and search panels.

SponsorFlow | Next.js, TypeScript, PostgreSQL, Drizzle ORM, NextAuth.js, Tailwind CSS | [GitHub](#)

Mar 2026

- Architected a full-stack sponsorship marketplace connecting event organizers and businesses using Next.js 16 and PostgreSQL.
- Implemented role-based authentication via NextAuth.js v5 and Drizzle ORM across 8 relational tables.
- Delivered tiered package management and proposal tracking with Tailwind CSS, Zod validation, and Zustand.

Calendar Application | Java, OOP, MVC Architecture, Data Structures, Algorithm Design, JUnit, TDD | [GitHub](#)

Dec 2025

- Built a multi-mode calendar app in Java supporting multiple time zones, recurring events, and GUI, CLI, headless modes.
- Applied OOP design patterns and MVC architecture to implement scheduling algorithms for time zone and recurrence handling.
- Achieved 90%+ code coverage with 60+ JUnit tests via TDD and optimized data structures across all execution modes.

StreamVault | R, MySQL, SQLite, SQL, Star Schema, RMarkdown | [GitHub](#)

Oct 2025

- Designed a cloud analytics datamart to analyze subscriber behavior and streaming trends from CSV and SQLite sources.
- Built an ETL/ELT pipeline in R and loaded a Kimball star schema on cloud MySQL with pre-aggregated fact tables.
- Engineered BI reports in RMarkdown and optimized SQL aggregations for drill-down queries across time, geography, and sport.